

The Limits of Verifiability in Communication

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The classic view is largely that verifiability improves communication

- ~ In many disclosure settings, it induces full disclosure, eliminating frictions from asymmetric info (Milgrom 81, Grossman 81, ...)
- ~ Whether communication should be verifiable is viewed as a matter of *technology*, not *design*

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Many institutions aim to make communication verifiable

In the United States, e.g.,

- ▶ The Securities Exchange Act prohibits firms from making untrue statements of material fact in connection with securities transactions
- ▶ Evidentiary rules restrict expert witness testimony to admissible and supportable evidence
- ▶ The FTC Act requires firms' communications to consumers to be truthful and backed by evidence

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- ~ Sender knows more than what her evidence can prove (i.e. evidence is noisy)
- ~ Sender and receiver have partially aligned objectives (a la CS'82)

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Main Result. If the sender's and receiver's objectives are

- ~ sufficiently misaligned, verifiability facilitates communication
- ~ sufficiently aligned, verifiability hinders communication

Main focus of the literature is on conditions for unraveling and its failures

Grossman 81, Milgrom 81, Dye 85, Jung and Kwon 88, Okuno-Fujiwara et al. 90, Seidmann and Winter 97, Mathis 08, Giovannoni and Seidmann 07, Hagenbach et al. 14

Disclosure when evidence is a noisy signal of private information

Fishman and Hagerty 90, Di Tillio et al. 21, Farina et al. 25

Other papers study how additional frictions affect the value of verifiability: costly disclosure (Jovanovic 82, Verrecchia 83), endogenous information acquisition (Matthews and Postlewaite 85, Shavell 94, Lizzeri 99), evidence design (DeMarzo et al. 19, Ali et al. 22), and agency frictions (Ben-Porath et al. 18)

model

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(S, π) is an **evidence structure** where

- ▶ $S = \{s_1, \dots, s_N\}$ and $s_1 < \dots < s_N$
- ▶ $\pi : \Theta \rightarrow \Delta(S)$

Evidence structure satisfies MLRP and “full support”, i.e., $\pi(s|\theta) > 0 \ \forall (\theta, s)$

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Evidence structure is *informative*, meaning that $\mathbb{E}(\theta \mid s_i) \neq \mathbb{E}(\theta \mid s_j)$ for some i, j

Timing:

1. A **sender** privately observes (θ, s) her “type”
2. She sends message to **receiver**, $m \in M \triangleq S \cup \{\circ\}$
3. Receiver observes m , updates her belief about θ , and takes action $a \in \mathbb{R}$

We will compare two opposite kinds of “evidence:”

Verifiable evidence: type- (θ, s) can either announce s or be silent, $m \in \{s, \circ\}$

- That is, sender can announce s only if s actually realized
- This is a **constraint** on sender's strategy $\sigma : \Theta \rightarrow \Delta(M)$ s.t. $\sigma(m|\theta, s) = 0$ if $m \notin \{s, \circ\}$

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Unverifiable evidence: type- (θ, s) can announce any s' or be silent, $m \in S \cup \{\circ\}$

- Sender can announce any s regardless of whether s actually realized

For the talks, let **payoffs** be

► For the receiver: $u(a, \theta) = -(a - \theta)^2$

► For the sender: $v(a, \theta) = -(a - \theta - b)^2$ $b \geq 0$ captures sender's bias

All results extend to larger class of payoffs a la Crawford-Sobel:

smooth, strictly concave in a ; single crossing in (a, θ) ; coincide at $b = 0$; and
higher b increases sender's preferred action

discussion

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Our game with **verifiable evidence** has two features that have rarely been combined (one exception is Haghtalab et al. '24)

1. Sender's evidence is a noisy signal of her private information
2. Sender and receiver's preferences are partially aligned

The *combination* of these two features drives our results

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Alphabet M is deliberately kept fixed $\rightsquigarrow$ only difference between **Unverifiable** and **Verifiable** is “*which types can send which messages*”

Many disclosure environments share these two features:

- ▶ Manager knows more about performance than formal metrics capture
- ▶ Loan officer knows more about borrower quality than financial history reveals
- ▶ An expert witness knows more than admissible evidence can convey

In all these cases,

- ▶ Sender knows more than what her evidence can prove
- ▶ Sender and receiver may be only partially aligned

outcomes and welfare

Some Basic Terminology

An **outcome** is a joint distribution of action and state $\lambda \in \Delta(A \times \Theta)$

An outcome is **uninformative** if action is independent of the state. Otherwise, it is **informative**

We say an outcome is **more informative** than another if it maximizes receiver expected payoff

We call a most-informative outcome **efficient** aka receiver-optimal

the disclosure game

The Disclosure Game

We begin by analyzing this sender-receiver game with **verifiable evidence**

Unlike in typical disclosure games, the set of equilibria is quite rich

Its characterization will lead to several insights, which are useful to understand main result later (comparison btw verifiable vs unverifiable evidence)

Does an analogue of the **unraveling result** holds in our setting?

- ▶ Clearly, receiver can't learn θ , but can she at least learn s ?
- ▶ I.e., does equilibrium exist where receiver learns the evidence s (i.e., the verifiable content of sender's private info)?

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We give such an outcome a name:

Definition. In the **full-evidence-disclosure (FED)** outcome, the receiver learns the realized signal s

An equilibrium is FED if it induces the FED outcome

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The **positive part** is in line with disclosure literature: when sender is very biased, unraveling occurs in (an) equilibrium

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The **negative part** is novel: when sender and receiver are sufficiently aligned, there is no eqm in which receiver learns the signal realization

To understand this failure, consider extreme case: $b = 0$ (common interest)

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Consider a sender's type (θ, s) , where $\theta = a_\circ$:

- ▶ Stick to eqm and send $m = s$, thus inducing action $\mathbb{E}(\theta \mid s)$ OR
- ▶ Deviate to $m = \circ$ and induce $a_\circ \in [0, 1]$

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Since evidence is nontrivial, $\exists \tilde{s}$ such that $\mathbb{E}(\theta \mid \tilde{s}) \neq a_\circ$, $\rightsquigarrow$ profitable deviation

This argument reflects a more general lesson:

- ▶ There is no way to assign an off-path interpretation to silence that is unattractive to **all** sender's types
- ▶ Any off-path action will “attract” some sender type, providing them with the incentives to **separate** from other types

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This **separating force** suggests sender may be able to transmit **more** information than that encoded in her verifiable evidence

Our next result confirms such intuition

Proposition 2

For all $b \geq 1$, full-evidence-disclosure (FED) equilibrium exists and is efficient

Conversely,

- ~ For all (F, S, π) , there is b sufficiently small such that the efficient equilibrium is **more informative** than the FED outcome

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Thus, when objectives are sufficiently aligned, unraveling is not only impossible, it is also inefficient

Idea: When b small, silence is not used to conceal unfavorable evidence, but to conceal misleading evidence

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Policy implication: when sender's bias is small, **mandated disclosure** can decrease the receiver's welfare

We characterized the set of equilibria in our game with verifiable evidence

- ▶ Differs qualitatively from the typical disclosure models

Note: these results are driven by the **combination** of the two key features

- ▶ Sender's evidence is a noisy signal of her private information
- ▶ Sender and receiver's preferences are partially aligned

In the absence **of either**, full-evidence-disclosure (FED) eqm exist and is efficient

verifiability vs unverifiability

Verifiable vs Unverifiable Communication

We compare efficient equilibria under verifiable and unverifiable communication

Which technology induces the most informative equilibria?

Note: in the standard disclosure model, this question is moot:

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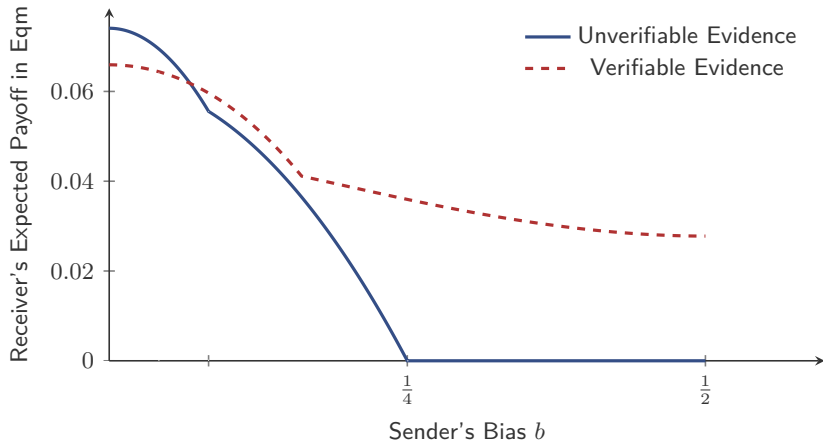
~ verifiability obviously dominates, since receiver fully learns θ

Proposition 3

Suppose evidence is informative.

- ▶ If $b \geq \max\{\frac{1}{2}, \bar{b}\}$, efficient eqm under verifiable evidence is **more informative** than efficient eqm under unverifiable evidence
- ▶ If b is sufficiently small, efficient eqm under verifiable evidence is **less informative** than efficient eqm under unverifiable evidence

An Example



Interpretation: Flexibility vs Credibility

When b is small,

- ▶ Under **verifiable** evidence, sender is constrained to use disciplined language,
 $m \in \{o, s\}$

She cannot freely choose the message that best conveys her private information

- ▶ By contrast, the flexibility of **unverifiable** evidence facilitates information transmission

Interpretation: Flexibility vs Credibility

When b is large,

- ▶ The flexibility of **unverifiable communication** damages the sender's credibility
- ▶ By contrast, the inflexibility of **verifiable communication** makes the sender more credible, because signals carry intrinsic information

Credibility and Flexibility

To better understand the tradeoff of Proposition 3, we ask

- ▶ What gives value to verifiable communication?

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Is the fact that verifiable signals carry information in their own right? credibility

- Unlike cheap talk, messages are not just labels without objective content

Or is the fact that verifiability constrains the sender's strategy? inflexibility

- These constraints curb imitation, which may facilitate communication

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To isolate first channel, we study our disclosure game but assume evidence is **uninformative**

In this case, any value created by verifiable communication must come from the constraints it imposes on the sender's strategy

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Result 2. Making evidence more Blackwell informative can make efficient equilibrium **less informative**

To isolate the role of inflexibility, we shut down the credibility channel

- ▶ We focus on the case in which the evidence structure is **uninformative**

As in cheap talk, signals are just labels with no informational content

Any value created by verifiable communication must therefore come from the constraints it imposes on the sender's strategy

We illustrate our result with uniform distribution

result holds for all F 's

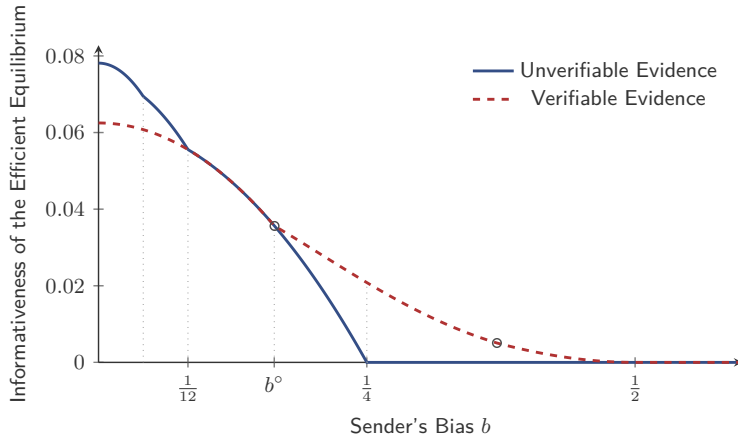
Proposition 4

Let F be uniform and (S, π) be uninformative. There is $b^\circ \in (\frac{1}{8}, \frac{1}{4})$ s.t.

- ▶ If $b \geq \frac{1}{2}$, efficient eqm is uninformative regardless of whether evidence is verifiable
- ▶ If $b \in (b^\circ, \frac{1}{2})$, efficient eqm is **more informative** when evidence is verifiable
- ▶ If $b \in (\frac{1}{12}, b^\circ]$, efficient eqm is equally informative regardless of whether evidence is verifiable
- ▶ If $b < \frac{1}{12}$, efficient eqm is more informative when evidence is unverifiable

Example

credibility and flexibility



Even when evidence is uninformative, verifiability can outperform unverifiable communication

- ▶ This shows that **inflexibility** alone can make verifiability valuable

What is the mechanism?

- ▶ Verifiable evidence, though uninformative, acts as an exogenous source of randomization
- ▶ This constraint disciplines deviations and allows the sender to sustain informative equilibria

This is akin to the role of noisy communication channels in cheap talk

Myerson '91, Krishna and Morgan '04, Blume et al. '07, Goltsman et al. '09

Does More Informative Evidence Help?

credibility and flexibility

But then, does the informativeness of evidence play any role at all?

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Remark 1

Suppose (S, π) is more Blackwell-informative than (S', π') .

- ▶ If $b = 0$ or $b \geq 1$, the efficient equilibrium under (S, π) is more informative than the efficient equilibrium under (S', π')
- ▶ For intermediate values of b , the opposite can happen

These failures are not knife-edge: Examples in the paper

interacting hard and soft info

Interacting Verifiable and Unverifiable Communication

So far, we have compared two polar regimes:

- ~ **Verifiable** communication: credible but constrained
- ~ **Unverifiable** communication: flexible but noncredible

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In many settings, communication often combines both

Our model provides a natural setting to study these interactions

- ~ Unlike in standard settings, there is scope for the two channels to complement each other

Interacting Verifiable and Unverifiable Communication

Everything as before, except now sender sends two messages (m^V, m^U)

$\sim m^V \in S \cup \{\circ\}$ is **verifiable**: type (θ, s) can choose only $m^V \in \{s, \circ\}$

$\sim m^U \in M^U$ is **unverifiable**: type (θ, s) can send any message in M^U

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~ $m^U \in M^U$ is **unverifiable**: type (θ, s) can send any message in M^U

We find that equilibria in this hybrid game can be less informative than equilibria in which only unverifiable evidence is available

This reinforces paper's broader theme:

- ~ The presence of verifiable evidence may limit flexibility and thus ability to transmit info
- ~ **Policy implication**: Mandating disclosure can reduce informativeness even when sender retains an unrestricted cheap-talk channel

conclusion

Conclusion

We study a standard communication model whose innovation consists of combining two simple features:

- ▶ The sender's evidence is a noisy signal of her private information
- ▶ The sender and receiver have partially aligned objectives

We use this model to qualify a central lesson of the disclosure literature: verifiability is not always a positive force in communication

Tradeoff: verifiability gives the sender credibility, but restricts her flexibility

We find that verifiability facilitates communication when objectives are sufficiently misaligned, but hinders communication when they are sufficiently aligned

The Limits of Verifiability in Communication

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Thank You

Basic observation: More can be better

Remark 1

The set of equilibrium outcomes of the hybrid game contains the union of the equilibrium outcome sets of the two benchmark.

- To replicate each benchmark, “babble” on the other dimension

Disclose-and-Refine

Class of equilibria where two channels are complement: **Disclose and Refine**

- ~ *Disclose*: Sender reveals s , the verifiable evidence
- ~ *Refine*: Conditional on s , cheap talk induces familiar monotone partitional structure

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In a DR equilibrium, the receiver learns the realization of the signal s with probability one

- ~ Counterpart of FED in the game with verifiable evidence only

Remark 2

If M^U is rich enough, a disclose-and-refine equilibrium exists for every $b \geq 0$

Why Disclose-and-Refine Exists?

Recall why FED can fail in the pure disclosure game:

- ~ All types with the same signal s induce the same action
- ~ But different θ -types with the same s want different actions
- ~ No single interpretation of silence can deter all deviations

Why Disclose-and-Refine Exists?

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- ~ No single interpretation of silence can deter all deviations

The cheap-talk channel fixes this problem:

- ~ After disclosing s , the sender can refine within the posterior F_s
- ~ This makes unraveling sustainable even when pure FED is not

But Disclosure Can Still Be Too Rigid

Disclose-and-refine is not necessarily the efficient equilibrium in the hybrid game

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Remark 3

The most-informative disclose-and-refine equilibrium can be strictly less informative than the efficient equilibrium under unverifiable evidence

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- ~ Flexibility can help info transmission. Verifiable evidence may limit that flexibility
- ~ **Policy implication:** Mandating disclosure can reduce informativeness even when sender retains an unrestricted cheap-talk channel